WORKSHEET 9.5

Free-Response Type: Functions Given by Graphs and Tables

Unit 9: Exam Strategy and Free-Response Types

Key ideas

- From a graph of f': f increases where $f' > 0$; f is concave up where f' is increasing; f'' is the slope of the graph.
- Values of f come from areas: $f(b) = f(a) + \int_a^b f'(x)\,dx$, with areas below the axis counted as negative.
- From a table: estimate a derivative with the two nearest points, and an integral with a Riemann or trapezoidal sum.
- $\int_a^b f'(x)\,dx = f(b) - f(a)$ exactly, whatever the spacing of the table.

Worked examples

Example 1 · A relative maximum

The three examples use the graph of f' printed at the top of the next page, with $f(-1) = 3$. At what value of x does f have a relative maximum?

Solution. f' changes from positive to negative where the graph crosses the axis going down.

$x = 0$.

Example 2 · A value from areas

Find $f(5)$.

Solution. $f(5) = 3 + \int_{-1}^{5} f'(x)\,dx = 3 + (1 - 1 - 2 + 4)$.

$f(5) = 5$.

Example 3 · Concavity

On what interval is the graph of f concave up?

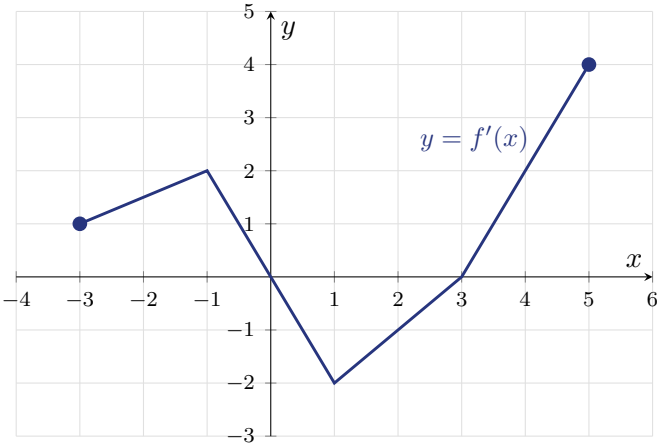
Solution. f' is increasing on $(1, 3)$ and on $(3, 5)$.

$1 < x < 5$.

AP© is a registered trademark of the College Board, which was not involved in the production of, and does not endorse, this material. © Integration Academy.

Practice

No calculator on this page. The graph of f', the derivative of f, consists of four line segments, as shown. It is given that $f(-1) = 3$. The worked examples on the first page and Questions 1 to 4 and 9 use this graph.



CORE

1. Find $f'(-2)$ and $f''(-2)$.

2. Find $f''(4)$.

3. Find $f(-3)$.

4. On what interval is f decreasing? Give a reason.

EXAM LEVEL

5. The table gives the temperature $H(t)$, in °C, of a liquid at selected times t minutes. Estimate $H'(5)$, with units.

t	0	4	6	10	14
$H(t)$	20	32	38	44	46

AP© is a registered trademark of the College Board, which was not involved in the production of, and does not endorse, this material. © Integration Academy.

6. Use a trapezoidal sum with the four subintervals of the table to estimate $\int_0^{14} H(t)\,dt$.

7. Use your answer to Question 6 to estimate the average temperature for $0 \le t \le 14$.

8. Evaluate $\int_0^{14} H'(t)\,dt$ and state its meaning.

Question 9 · Free-response practice

Use the graph of f' at the top of the page, with $f(-1) = 3$.

(a) Find the absolute minimum value of f on $[-3, 5]$. Justify your answer.

(b) Find the x-coordinate of each point of inflection of the graph of f. Give a reason.

(c) Let $h(x) = x\,f(x)$. Find $h'(1)$.

AP© is a registered trademark of the College Board, which was not involved in the production of, and does not endorse, this material. © Integration Academy.

Worked solutions

1. $f'(-2) = \dfrac{3}{2}$ and $f''(-2) = \dfrac{1}{2}$.
2. $f''(4) = 2$.
3. $f(-3) = 3 - \displaystyle\int_{-3}^{-1} f'(x)\,dx = 3 - 3 = 0$.
4. $0 < x < 3$, because $f'(x) < 0$ there.
5. $\dfrac{38 - 32}{6 - 4} = 3$ °C per minute.
6. $\dfrac{4(52)}{2} + \dfrac{2(70)}{2} + \dfrac{4(82)}{2} + \dfrac{4(90)}{2} = 104 + 70 + 164 + 180 = 518$.
7. $\dfrac{518}{14} = 37$ °C.
8. $H(14) - H(0) = 26$: the temperature rises 26 °C over the 14 minutes.
9. (a) Candidates are the endpoints and $x = 3$, where f' changes from negative to positive. $f(-3) = 0$, $f(3) = 3 + 1 - 1 - 2 = 1$ and $f(5) = 5$. The absolute minimum value is 0, at $x = -3$.
   (b) $x = -1$ and $x = 1$: f' changes from increasing to decreasing at $x = -1$ and from decreasing to increasing at $x = 1$.
   (c) $h'(x) = f(x) + x\,f'(x)$, and $f(1) = 3 + 1 - 1 = 3$, $f'(1) = -2$: $h'(1) = 3 - 2 = 1$.

AP© is a registered trademark of the College Board, which was not involved in the production of, and does not endorse, this material. © Integration Academy.